

Meghana Aparna Sistla

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Research Interests

- **Primary Interests:** AI for Code and Formal Reasoning, Program Analysis, Formal Methods.
- **Other Interests:** Network Verification, Quantum Computing.

Education

Ph.D., The University of Texas at Austin Aug 21 - May 26 (Expected)
Computer Science, Advisor: Swarat Chaudhuri
GPA: 4.0/4.0
Thesis: Succinct Boolean function representations for simulation and verification.

B.Tech + M.Tech, Indian Institute of Technology Madras Aug 14 - May 19
Computer Science and Engineering, Advisor: V. Krishna Nandivada
GPA: 9.41/10
Thesis: Graph Coloring using GPUs [↗](#).

Selected Publications & Preprints

- **Meghana A. Sistla**, Gogul Balakrishnan, Pat Rondon, José Cambronero, Michele Tufano, and Satish Chandra. Towards Verified Code Reasoning by LLMs. **Under Review, 2025** [↗](#)
- A. Thakur, J. Lee, G. Tsoukalas, **M. Sistla**, M. Zhao, S. Zetsche, G. Durrett, Y. Yue, and S. Chaudhuri. CLEVER: A Curated Benchmark for Formally Verified Code Generation. Neural Information Processing Sys., **NeurIPS 2025** [↗](#)
- **Meghana A. Sistla**, Swarat Chaudhuri, and Thomas Reps. Weighted Context-Free-Language Ordered Binary Decision Diagrams. Object-Oriented Programming, Systems, Languages, and Applications (**OOPSLA**), **2024** [↗](#)
- **Meghana A. Sistla**, Swarat Chaudhuri, and Thomas Reps. CFLOBDDs: Context-Free-Language Ordered Binary Decision Diagrams. ACM Trans. on Program. Langs. and Systems (**TOPLAS**), **2024**. Jour.-first at **PLDI 2024** [↗](#)
- **Meghana A. Sistla**, Swarat Chaudhuri, and Thomas Reps. Symbolic Quantum Simulation with Quasimodo. International Conference on Computer-Aided Verification (**CAV**), **2023** [↗](#)

Professional Experience

Google, DevAI Research Team May 25 - Aug 25
PhD Intern, Host: Satish Chandra, Co-host: Gogul Balakrishnan (Google DeepMind)
Sunnyvale, USA
Paper: **Towards Verified Code Reasoning by LLMs**. Under Review.

- Designed and implemented a mechanism to **verify the LLM output** for program equivalence questions.
- Used in-context learning with an agent system using Gemini 2.5 Pro to convert LLM textual output to a machine checkable format and verified using formal techniques.
- Caught *LLM hallucinations* on 6/15 program-equivalence tasks and verified the output of the remaining 9/15.

Google, P4-Based Automated Reasoning Group May 24 - Aug 24
PhD Intern, Host: Ali Kheradmand, Co-host: Steffen Smolka
Sunnyvale, USA

- Faced with a challenge of optimizing a symbolic execution tool that generate tests for hardware switch testing.
- Leveraged program analysis techniques and Z3 internals to improve tool performance and extend test coverage.
- Achieved up to **800x speedup**, reducing runtime from 2 hrs to < 10 secs, and deployed immediately to production.
- Presented the work at P4 Workshop 2024 [↗](#).
- Awarded *two peer bonuses* for the impact of the work.

Google India Oct 20 - Aug 21
Software Engineer II, Manager: Ankur Lahoti
Bangalore, India

- Recovered **\$1.5M** by modeling untracked conversions for the Google Ads team.

Microsoft, India Development Center Jul 19 - Oct 20
Software Engineer, Manager: Saibaba Konduru
Hyderabad, India

- Enhanced **performance and reliability** of IaaS VM deployments in the Azure Compute team.

Selected Research Projects

AI for Code/Math Projects

Verified Autoformalization using LLMs

Oct 25 - Present

Research (Project Co-Lead), Prof. Swarat Chaudhuri

UT Austin

- Researching an important problem of auto-formalization – transforming informal mathematical propositions into verifiable formal representations – in LEAN.
- Developing evaluation framework and high-quality dataset for benchmarking theorem formalization models.
- Experimented with LLM fine-tuning techniques to improve model accuracy.

CLEVER: A Curated Benchmark for Formally Verified Code Generation

Mar 25 - May 25

Research (Project Contributor), Prof. Swarat Chaudhuri

UT Austin

- Created CLEVER, a benchmark of **161 problems** for end-to-end verified code generation in LEAN.
- Includes dual tasks: specifications aligned with ground truth, and synthesizing LEAN implementations with proofs.
- Benchmarked multiple large language models and baselines on both tasks using the *pass@k* metric; showing no existing model achieves full formal correctness.

Formal Methods, PL projects

Succinct function representations for formal simulation and verification

Aug 21 - Present

Research (Project Lead), Prof. Swarat Chaudhuri, Thomas Reps

UT Austin

Context-Free-Language Ordered Binary Decision Diagrams

– Paper: M. Sistla, S. Chaudhuri, T. Reps. [↗](#)

- Developed theorems and algorithms for a new data-structure – CFLOBDDs – for efficient function representations.
- Applied to the domain of quantum simulation where exponentially large matrices and vectors need to be represented.
- Proved theoretically and showed empirically that CFLOBDDs provide **exponential compression** over BDDs.
- CFLOBDDs provide upto **1024x speedup** over BDDs on some quantum benchmarks; with 128x for Grover's algorithm.

Weighted Context-Free-Language Ordered Binary Decision Diagrams

– Paper: M. Sistla, S. Chaudhuri, T. Reps. [↗](#)

- Designed Weighted CFLOBDDs – a weighted version of CFLOBDDs, to overcome challenges faced by CFLOBDDs.
- Developed algorithms and proved exponential compression over Weighted BDDs and CFLOBDDs.
- Applied to quantum simulation with WCFLOBDDs performing upto **64x speedup** over WBDDs and **128x speedup** over WCFLOBDDs – best of both worlds.

Quantum Simulation with Quasimodo

– Paper: M. Sistla, S. Chaudhuri, T. Reps. [↗](#)

- Developed an easily extensible tool – Quasimodo – for quantum simulation supporting quantum operations; with easy support for many data structures (CFLOBDDs, BDDs, WCFLOBDDs, and WBDDs).
- Designed Quasimodo to be able to easily extend to any data-structure of choice.

Academic and Scholastic Achievements

- PC Member for Artifact Evaluation for OOPSLA 2025, POPL 2025, PLDI 2024, POPL 2024.
- Invited External Reviewer for ICALP 2024; sub-reviewer for LICS 2022.
- Awarded **Chair's Strategic Excellence Fellowship** at UT Austin for competitive admits for academic year 2021-2022
- Awarded **B.Lakshmi Ravi Best Dual Degree Project Award** for the best thesis in the Computer Science department.
- Awarded S.N. Bose Scholarship by the Govt. of India to undertake a research internship for the summer of 2018.
- Secured an All India Rank of 219 in Jee Advanced 2014 among 2,00,000 students.

Course Work & Skills

- Relevant Coursework** - Model Checking, Program Synthesis, Verification of Distributed Systems, Program Analysis. Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Processing, Computer Networks.
- Skills** - C++, Python, PyTorch, Numpy, LEAN, Z3, CUDA, C, Java, JavaScript, Django, Git, L^AT_EX, SQL.